

ESTEFANIA TORREJÓN

COMPUTATIONAL BIOLOGIST - PHD STUDENT

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SUMMARY

With 5 years' experience in the area of computational biology, I specialise in developing novel computational methods and bioinformatic approaches for complex omics data analysis. My work aims to uncover insights from large-scale biomedical datasets and contribute to innovative therapeutic strategies. Although I now spend most of my time in front of the computer, my previous research experience in Lima (Peru) and Lisbon (Portugal) equipped me with valuable wet-lab skills that have supported impactful, interdisciplinary research projects. I am proficient in R and Python for data analysis and software development, as well as C for low-level algorithm development. Beyond my scientific endeavours, I am a key contributor of the International Society of Computational Biology (ISCB) Student Council and the Peruvian Society of Computational Biology (SPBBC) and have led many of their outreach initiatives.

RESEARCH AND WORK EXPERIENCE

iNOVA4Health PhD fellow @ NOVA Medical School.

Since January, 2022

Lisbon, Portugal

- Omics (big biological data) analysis and its integration with complex medical data.
- Development of an R package for the prediction of the extracellular vesicles effect in their target tissues: EV-Net (see more info at <https://torrejonia.github.io/EV-Net/>).
- Wrote a successful grant application for an EMBO short scientific exchange.
- Extensive knowledge and analysis of relevant scientific and clinical literature
- Collaborated with the Curry Cabral Hospital staff and had direct contact with the cohort of patients.
- Collaborated with different research groups, both in Europe and in the USA, showing ability to thrive in changing work environments.
- Isolation of extracellular vesicles.
- **Tech:** R, Python, JSON, Markdown

Research fellow @ the Bee keeping and biodiversity Unit.

July, 2022 – January, 2023

Instituto Nacional de Investigação Agrária e Veterinária (INIAV). Lisbon, Portugal

- Monitoring of *Vespa velutina* as an invasive species.
- Kept the monitoring network up to date through the ArcGIS (SIG) platform.
- SQL queries and automation of emails using Python Scripts.
- Data management.
- **Tech:** SQL, Python, ArcGIS

Research fellow @ the Structural Biology of Drug Discovery Unit – Merck Satellite Lab. iBET. Lisbon, Portugal

September, 2020 – June, 2022

- Developed and executed a scientific project from beginning to end.
- Kept the monitoring network up to date through the ArcGIS (SIG) platform.
- Developed protocols for CETSA, CETSA-MS, Mass spectrometry based untargeted proteomics
- Established a workflow for thermal proteome profiling analysis and published a manuscript exhibiting the results of the project.
- Developed R-based pipelines for thermal proteome profiling analysis using open-source libraries, saving the research group approximately €20,000 in proprietary software costs.
- **Tech:** R, OpenMS, GraphPad, MaxQuant

Researcher @ the National Reference Laboratory of Bacterial Vector-borne Diseases. National Center of Public Health. Instituto Nacional de Salud. Lima, Peru **January, 2016 – September, 2017**

- Serological diagnosis of “Carrion Disease”.
- Identified Bartonella sp. DNA in patients’ plasma and Lutzomyia vectors.
- Developed workshops for health professionals coming to different regions of Peru, with the aim of increasing their portfolio of Carrion disease diagnosis technologies.

EDUCATION

Software Engineering with a focus on AI @ 42 Lisboa

- Ongoing

PhD in Health Sciences @ NOVA Medical School. Universidad NOVA de Lisboa. Lisbon, Portugal.

- Ongoing

MSc in Biomedical Sciences @ the Institute of Hygiene and Tropical Medicine. Universidad NOVA de Lisboa. Lisbon, Portugal.

- Graduated: 2020

BSc in Biological Sciences @ the Universidad Nacional Mayor de San Marcos. Lima, Peru.

- Graduated: 2015

SCIENCE COMMUNICATION, LEADERSHIP AND OTHER ROLES

- **Head of Advertisement. Organizing Committee of the Latin American ISCB Conference 2026 (LA-SCS2024)** to be carried out in Lima, Peru. **Since 2026**
<https://www.iscb.org/latam2026/organizers/>
- **Supervisor of two Master’s students** **Since 2026**
- **Chair of the Latin American ISCB Student Council Symposium (LA-SCS2024)** carried out in Medellin, Colombia. <https://iscbsc.org/la-scs2024/> **2024**
- **Director of Foreign Affairs of the Peruvian Society of Bioinformatics and Computational Biology.** <https://bioinformatica.pe/organizacion/> **Since 2024**
- **Member of the Student Council of the International Society of Computational Biology (ISCB)** actively involved in organizing the ISCB Student Council Symposium 2023 and 2025. **Since 2023**
- **23 Poster presentations** and **15 Oral presentations** in national and international conferences.

AWARDS

- Selected for the **rOpenSci Champions Program2026** **2026**
- **Best Poster Award in the category “tool” granted to “EV-Net”** by the ISCB's NetBio CoSI at the ISMB/ECCB 2025 conference in Liverpool, UK. **2025**
- **Travel fellowship** granted by the Twinning Project Extracellular Vesicles in Clinical Applications (EVCA) to attend the ISEV2025 conference in Austria. **2025**
- **EMBO scientific exchange grant** for a visit to the Marseille Medical Genetics Research Center’s System Biomedicine lab (3 months). **2024**
- **Travel fellowship and registration fee waiver** for the ISMB/ECCB 2023. **2023**
- **PhD fellowship** “iNOVA4Health – Translational Medicine programme”. **2022**
- **Best poster Award** at the “BioTicks2019” conference. **2019**
- **Scientific exchange fellowship** “Movilización Nacional e Internacional en Ciencia, Tecnología e Innovación 2016-04” by the Peruvian Foundation for Science and Technology **2017**
- **Excellence Award (Valedictorian Title)** for achieving the top rank among all students throughout my high school years **2008**

PUBLICATIONS

- **EV-Net: A computational framework to model extracellular vesicles-mediated communication**, Torrejón, E., [...], Baudot, A., Machado de Oliveira, R. *bioRxiv* (pre-print).
- **Molecular and phylogenetic insights into Polychromophilus diversity in bat flies (Diptera: Nycteribiidae) from Portugal**. André, M. R., de Mello, V. V. C., Barbier, E., Mariano, L. C., Rodrigues, L., Torrejón, E., [...] & Sanches, G. S. *Parasitology International*, 2026. *IF*₂₀₂₄ 1.9
- **Molecular Survey of Anaplasmataceae Agents, Rickettsia spp., Bartonella spp., and Piroplasmids in Ectoparasites from Cave-Dwelling Bats in Mainland Portugal**. Sanches, G. S., Rodrigues, L., Torrejón, E., [...] & André, M. R. *Pathogens*, 2025. *IF*₂₀₂₄ 3.3
- **Adapting beyond borders: insights from the 19th student council symposium (SCS2023), the first hybrid ISCB student council global event**. Al Sium, S. M., Torrejón, [...] & Olguín-Orellana, G. J. *Bioinformatics Advances*, 2024. *IF*₂₀₂₄ 2.8
- **Aldehyde Dehydrogenase 2 (ALDH2): a novel sorafenib target in hepatocellular carcinoma unraveled by the proteome-wide cellular thermal shift assay**. Ferreira, I. C. + , Torrejón, E., [...], Bortoluzzi, A. & Rebelo, S. P. *SLAS Discovery*, 2024. *IF*₂₀₂₄ 2.7
+ These authors contributed equally.
- **Proteomic Profiling of Plasma and Gut-Derived Extracellular Vesicles in Obesity**. Pereira, P. B., Torrejón, E., [...], Macedo M. P. & Machado de Oliveira, R. *Nutrients*, 2024. *IF*₂₀₂₄ 5
- **Molecular survey of Bartonella species in stray cats and dogs, humans, and questing ticks from Portugal**. Torrejón, E., [...], Domingos, A., & Antunes, S. *Pathogens*, 2022. *IF*₂₀₂₂ 3.4